

JEFFREY TSENG

tseng94@purdue.edu | jtseng.org
Purdue University

PROFESSIONAL SUMMARY

Full stack developer with experience in defensive backend radar systems, autonomous racing, machine learning, and website and app development, seeking a 2026 software engineering internship.

EDUCATION

Purdue University, West Lafayette, IN Graduation: May 2029
Bachelor of Science in Computer Science and Mathematics GPA: 4.0/4.0
Relevant Classes: CS390: Web Applications Programming, CS193: Computer Science Tools, CS180: Problem Solving & Object-Oriented Programming, MA261: Multivariable Calculus

RELEVANT EXPERIENCE

IEEE Autonomous Racing Purdue University
AI Perception Team August 2025 – Present

- Developed **AI perception pipelines** for autonomous racing vehicles using **lidar** and **stereo cameras**, creating **depth estimations accurate up to 35 meters**
- Implemented **AI/ML algorithms** for object detection and path planning in **ROS2**
- **Collaborated** across submodules to integrate software for **real-time deployment**

MIT Beaver Works Cambridge, MA
Synthetic Aperture Radar Developer July 2024 – August 2024

- Developed radar imaging **back-projection algorithms in Python and C++** using NumPy, SciPy, and Matlab to process radar data and generate images, processing radar data enabling **high-resolution images** of 2.6 in. soda cans across 10 by 10 meters
- Developed Synthetic Aperture Radar (SAR) applications in **collaboration with MIT Lincoln Labs** and the Department of Defense to **identify foreign objects through foliage**
- Built an F550 Hexacopter, with a **PulsOn 440 radar** with a **Raspberry Pi via socket programming** for **live data transmission**

Idiopathic Pulmonary Fibrosis Research Stanford University, CA
Researcher June 2023 – October 2023

- Conducted wet lab experiments on **human lung fibroblasts**, focusing on Idiopathic Pulmonary Fibrosis and the effect of pirfenidone on serum-stimulated human lung fibroblasts
- Performed **bulk and single-cell RNA sequencing** after extensive training in R, **applying data science to biological research**
- **Collaborated with Stanford researchers** to explore findings and **contribute to ongoing research** on pulmonary diseases

LEADERSHIP AND INVOLVEMENTS

Google Developer Groups Purdue University
Technical Teaching Assistant August 2025 – Present

- Designed and deployed take-home projects for **300+ students teaching React, Google Cloud, Vertex AI, & Tensorflow**
- **Mentored 150+** developer role students, providing technical assistance on **Flutter, Firebase, and full-stack projects**
- **Selected as one of two technical mentors** from over 250+ applicants

PERSONAL PROJECTS

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- Architected and deployed a dynamic, **full-stack personal portfolio website** to showcase skills and professional experience
- Engineered the frontend with **React and TypeScript**, creating a responsive, mobile-first user interface with **modern design** and a seamless user experience
- Developed a **robust backend using Node.js** to handle API requests and serve project data, ensuring efficient performance
- Implemented a **clean, component-based architecture** to enhance code maintainability, reusability, and future scalability
- Designed and built site from the **ground up**, from initial concept and design to final deployment

TECHNICAL SKILLS

Languages: Python, Java, HTML, CSS, C++, C, TypeScript

Libraries and Tools: NumPy, SciPy, Matlab, Gazebo, ROS, Node, React, Tailwind

References Available Upon Request